

ContractIQ-SLM

Parameter-Efficient Domain Adaptation for Grounded Contract Intelligence

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[GitHub Repository](#) | [5-Minute Walkthrough Video](#)

2,856 SFT train examples	0.2825% trainable params	0.9101 final test F1	98.34% grounded evidence
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1. Problem & Domain

Task. Given a target legal clause, short clause guidance, and a contract passage, the model decides whether the clause is present and returns supporting evidence in strict JSON. The goal is correct classification plus evidence grounded in the supplied text.

Data. CUAD v1 contains 510 commercial contracts and 41 clause categories. I used eight categories covering liability, audit, termination, exclusivity, renewal, change of control, uncapped liability, and notice periods.

Why fine-tune?	The baseline was precise (0.9429) but recall was only 0.3976, so many valid clauses were missed. Prompting did not fix this. RAG was not the main need because the relevant passage was already provided.
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2. Approach & Algorithm Decisions

SFT data preparation - Contract-level split: 357 contracts / 2,856 examples for training, 76 / 608 for validation, and 77 / 616 for final testing. Test contracts remain unseen during training. - Each example contains target clause, clause guidance, contract passage, and expected JSON. Positive passages contain annotated evidence; negatives use realistic same-contract legal text where possible. - Max sequence length: 3,072 tokens. Loss is applied only to the assistant answer.	Model and adaptation choices Base model. Qwen2.5-7B-Instruct was selected for compact size, instruction following, structured output, and PEFT support. Mistral-7B-Instruct and Qwen2.5-3B-Instruct were design alternatives, not separate fine-tuning runs. Why DoRA. DoRA keeps the base model frozen while adapting weight magnitude and direction. Only 21.6M parameters (0.2825%) were trainable. Why BF16 DoRA over QLoRA here. QLoRA is ideal when memory is tight. With an A100 80GB available, BF16 avoided 4-bit quantization, kept the setup simpler, and removed quantization as an extra variable. Full fine-tuning would add unnecessary compute and forgetting risk.
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Final training configuration

Rank / alpha	8 / 16	LR / epochs	1e-4 / 2
Dropout / precision	0.05 / BF16	Effective batch	8

3. Results & Error Analysis

Validation capability lift. On the same 608-example validation set, recall improved from 0.3976 to 0.9217 and F1 from 0.5593 to 0.9301, while precision stayed high.

Model	Accuracy	Precision	Recall	F1
Baseline Qwen	0.8289	0.9429	0.3976	0.5593
DoRA fine-tuned	0.9622	0.9387	0.9217	0.9301

Same-prompt example

Gold evidence	Either party may terminate this Agreement without cause at any time effective upon thirty (30) days' written notice.
Baseline Qwen	present=false, evidence=null
DoRA fine-tuned	present=true, evidence=the exact supporting sentence above

Final untouched test. After fixing the configuration on validation, I evaluated once on 616 examples from 77 unseen contracts. No test-time retuning was done.

0.9448 Accuracy	0.9101 F1	0.9351 Macro F1	0.9834 Grounded evidence
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Error analysis • 24 false negatives and 10 false positives. The weakest category was Notice Period To Terminate Renewal (recall 0.625). • 3 unsupported-evidence cases (1.66%), 3 invalid JSON outputs, 4 schema-invalid outputs, and 3 responses at the 256-token generation limit.	Generalization checks • Contract-level splitting kept all 77 test contracts unseen during training; test F1 was 0.9101 and evidence grounding was 0.9834. • A 15-prompt general-capability sanity check gave 15/15 for both base and fine-tuned models, with 0 observed regressions. This is a limited diagnostic, not a full forgetting benchmark.
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4. Production & Limitations

Production path • Serve the merged model behind a simple backend API and document-review interface. • For full contracts, add document ingestion and retrieval/passage selection before inference. • Validate JSON schema and confirm evidence is present in the source passage; route low-confidence or unsupported cases for review. • Average test latency was about 1.04 s/example on the A100. Lower-cost serving should separately test quantization and optimized runtimes.	Limitations and next steps • The current experiment assumes a relevant passage is already available. • Only 8 of CUAD's 41 clause categories were used. • Next steps: full-document retrieval, broader clause coverage, external contract evaluation, and a larger general-capability benchmark.
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Takeaway. A small parameter-efficient update substantially improved contract-clause recall and grounded evidence extraction, while keeping the model compact and preserving the general capabilities tested in this study.